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History of Topology

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Abstract

In[1] this paper we will give a brief overview of the history of topology through two lines of historical development, and how their synthesis came about:

1. *Geometric-combinatorial* topology
2. *General*[2] topology

Geometric-combinatorial topology arises from the study of certain properties of polyhedra. General topology is a generalization of analysis to arbitrary sets and develops together with set theory. The first line begins a couple of centuries before the second and was not fully understood as a separate mathematical discipline, distinct from geometry. As such, it initially

[1]This document was originally written in Bosnian language, for a course in Bosnian language. This is the English language version. The translation was done by an LLM, although some effort into that process has been put, the translation was not carefully checked. The translation may also have compromised the typesetting and aesthetics of the document. Worth noting here is that LLMs were not used for the production of the writing of the original work (in Bosnian), except for literature search.

[2]also known as "point-set topology"

had little direct influence on the second line of development, and on the emergence of "topology" as a formal mathematical discipline in the 19th century. But, as we will see, it will soon become clear that the two lines converge toward the same discipline, namely the topology we know today. Finally, we will look at the influence of topology on modern mathematics from the middle of the last century to the present.

Contents

1 Topology and Geometry **3**
- 1.1 Introduction . . . 3
- 1.2 Descartes's and Euler's Formulas for Polyhedra . . . 3
- 1.3 Classification of Surfaces . . . 5
- 1.4 Surfaces Equipped with Geometry . . . 8
- 1.5 The Fundamental Group . . . 12
- 1.6 The Poincaré Conjecture . . . 14
- 1.7 Transition . . . 18

2 General Topology **19**
- 2.1 Georg Cantor and the Beginnings of Point-Set Topology . . . 19
- 2.2 Fréchet and Metric Spaces . . . 20
- 2.3 Felix Hausdorff and Neighborhoods . . . 22
- 2.4 Transition . . . 26

3 Synthesis **27**
- 3.1 Dimension . . . 27
- 3.2 Peano, Jordan, and Their Curves . . . 29
- 3.3 Hurwitz and His Program . . . 31
- 3.4 Hausdorff and the Initiation of the Synthesis . . . 33
- 3.5 Brouwer's Contribution, the Return to Dimension, and the Beginnings of Algebraic Topology . . . 33
- 3.6 On Manifolds . . . 37

4 Influence on Modern Mathematics **39**
- 4.1 Algebraic Topology and Category Theory . . . 39
- 4.2 Brouwer's Intuitionism and Topology . . . 43

References **46**

1 Topology and Geometry

Historically, the development of geometric topology in this section mostly follows Stillwell's exposition [Sti10], with small adaptations and translations. We believe this is an excellent work, thorough enough for this part, while in the other parts the literature will be considerably richer.

1.1 Introduction

The theory of general topology, although important for set theory and analysis, is not geometric in nature by itself, so we can regard as geometric topology that branch of topology dealing with continuous transformations on $\mathbb{R}^n$ or on subsets thereof. Below, when we say "geometric topology"[3], "geometric" will serve as a reminder to differentiate this branch of topology from the others, but it is important to note that this geometry is still of a discrete and combinatorial nature. Namely, geometric quantities such as length, angles, and curvature are subject to continuous change and cannot be invariant under continuous transformations, so topology does not concern itself with them. The relevant quantities in topology are those invariant under continuous transformations, such as the number of "parts" of a figure or the number of "holes" (genus) in a figure. It will turn out that the combinatorial structures of topology have their reflection in the combinatorial structures of ordinary geometry, such as polyhedra and tessellations. For example, a cube and a sphere are topologically the same objects, and the Euler characteristic is a topological invariant, so the cube and the sphere must have the same Euler characteristic ($E_K = E_S$). Indeed, since the Euler characteristic of the cube is $E_K = 8 - 12 + 6 = 2$, it must also be that $E_S = 2$. On the other hand, we can obtain the torus by the topological technique of gluing a square (Figure 3), so the torus has 2 edges, one vertex, and one face, hence the Euler characteristic of the torus is $E_T = 1 - 2 + 1 = 0$. In the history of geometric topology we will deal only with the topology of surfaces, whose historical starting point is precisely the theory of polyhedra [Sti10, p. 468].

1.2 Descartes's and Euler's Formulas for Polyhedra

It seems that the first topological property of polyhedra was discovered by Descartes in 1630; although the original document has not survived, its con-

[3] We note this because today there exists an entire discipline called geometric topology, which is not what is meant here.

tents are known because Leibniz made a copy of Descartes's document in 1676. The same property was discovered independently of Descartes by Euler in 1752, in explicit form, and this property is now known as the Euler characteristic. Below we give the original definition of the Euler characteristic for polyhedra.

Definition. Let P be a polyhedron with V vertices, S edges, and F faces; then the Euler characteristic of the polyhedron P is given by

$$E_P = V - S + L$$

However, Euler showed that $E_P = 2$ for all convex polyhedra, a result now known as *Euler's formula for polyhedra.* This result had already been implicit in Descartes's formulas:

Proposition. Let B be the number of angles of a convex polyhedron P; then

$$B = 2L + 2V - 4, B = 2S.$$

The relation $B = 2S$ follows from the fact that each edge of the polyhedron forms two angles.

Note that all the quantities B, L, V, S are used in a combinatorial sense, namely denoting the number of angles, faces, vertices, and edges, respectively, and as such are not geometric quantities. Given that we are not concerned with the geometric properties of angles, edges, etc., the angle in Descartes's context is a topological concept, just as the concept of edge in Euler's formula is a topological concept.

For these reasons, Descartes's and Euler's results are classified as results of topology, even though topology did not yet exist as a mathematical discipline at that time. The question of whether Descartes or Euler was the "inventor" of topology is open and a matter of debate; for an overview of this topic see Federico (1982). J. Stillwell argues that neither of them understood these formulas in a fully topological sense, i.e., in their full topological generality, because they used angle measure in their proofs and did not realize that vertices, edges, and faces are meaningful notions on any surface: edges need not be straight (uncurved), and faces need not be flat (uncurved). Other early proofs also rely on angle measure and other strictly geometric quantities. For example, Legendre's proof (1794) assumes that the polyhedron can be projected onto a sphere, and then uses certain relations between the angles and areas of spherical polygons. Poincaré was probably the first who can be said with certainty to have understood $V - S + L$ in a purely topological sense (1895). He generalized the Euler characteristic to n-dimensional figures. But in the case of polyhedra, his key observation was the

following: a vertex divides an edge into two edges, and an edge divides a face into two faces; it follows that any subdivision of the edges or faces of a polyhedron leaves $V - S + L$ unchanged: if a new vertex is added on an edge, V and S both increase by 1; if a new edge is added on a face, then S and L increase by 1. A similar result would hold if we considered the removal of vertices and edges, etc. [Sti10, pp. 469–470].

1.3 Classification of Surfaces

Between 1850 and 1880, several different lines of research created the need for a topological classification of surfaces:

- One line was "downstream" of Euler, namely the classification of polyhedra
- The second line developed through Riemann (1851, 1857), namely the representation of algebraic curves by Riemann surfaces
- Connected to the second line was also the problem of classifying the symmetry groups of tessellations, considered by Poincaré (1882) and Felix Klein (1882) (more on this later)
- Finally, there was also the problem of classifying smooth closed surfaces, by Möbius (1863)

These various efforts began to converge when it became clear that in every case a surface could be divided into faces by means of edges (not necessarily edges that are straight line segments), so that one could then speak of a generalized polyhedron, and these are what was traditionally called a closed surface, today described in topology as "compact spaces without boundary".

It turned out that all polyhedra have an invariant Euler characteristic, not only objects homeomorphic to the sphere, and not only those with "flat" edges and "flat" faces. Mathematicians quickly realized, among them Riemann (1851) and Jordan (1866), that any closed surface is determined by its Euler characteristic, up to homeomorphism. They also realized that different Euler characteristics are achieved on surfaces which they called normal forms (Figure 1), discovered by Möbius (1863). For example, all surfaces homeomorphic to the sphere have the same Euler characteristic, and similarly for all surfaces homeomorphic to the torus (they have characteristic 0), or to a torus with several holes (they have characteristic $2 - 2g$, where g is the number of holes[4]).

[4]We also call the number of holes the genus, hence the letter g.

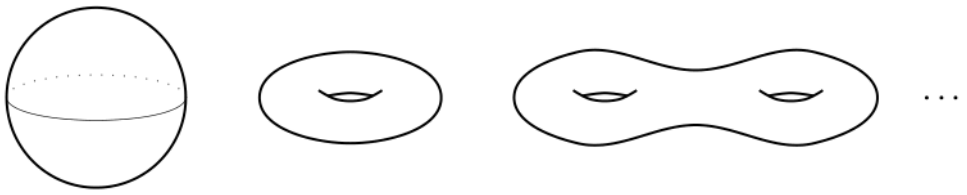


Figure 1: Closed orientable surfaces in normal form [Sti10, p. 472]

They conjectured that these forms differ topologically because of the number of holes, and an ambitious goal soon emerged: to show that any closed surface is homeomorphic to one of the closed orientable surfaces. But the assumptions Riemann and Möbius worked with were too specific for a purely topological proof. Riemann assumed the surface was a Riemann surface, while Möbius assumed the surface was embedded in $\mathbb{R}^3$. Moreover, their proofs contained a hidden assumption of orientability of the surface. All these assumptions affected the generality of the result, so the search for a proof independent of these assumptions continued. In 1907, starting from an axiomatization of the generalized polyhedron, Dehn and Heegaard gave a rigorous proof. It would turn out that the conjectures were on the right track; indeed, all closed orientable surfaces can be homeomorphically mapped to one of the surfaces in the figure.

However, it will turn out that besides orientable surfaces there also exist non-orientable surfaces such as the Möbius strip (Figure 2), independently discovered by Möbius and Listing in 1858.

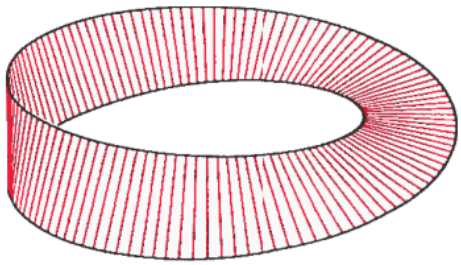


Figure 2: The Möbius strip, an example of a non-orientable surface [Wee20, p. 230]

A non-orientable surface can be defined as a surface containing a Möbius

strip. Non-orientable surfaces cannot be Riemann surfaces, nor can they lie in $\mathbb{R}^3$ without self-intersection; nevertheless, among them are some important surfaces, such as the projective plane. Non-orientable surfaces are likewise determined up to homeomorphism by their Euler characteristic.

One of Felix Klein's great contributions to topology was that, in 1882, he succeeded in assigning a polyhedral structure to Möbius's normal forms. Namely, these are minimal subdivisions of a given surface, with only a single face, and, except for the sphere, with only a single vertex. When Klein's subdivision of the surface is carried out, i.e., when the surface is cut along the edges and sides of the subdivision, one obtains the so-called fundamental polygon (Figures 3....), from which the surface can be reconstructed, namely by identifying ("gluing") the corresponding edges of the resulting polygon.

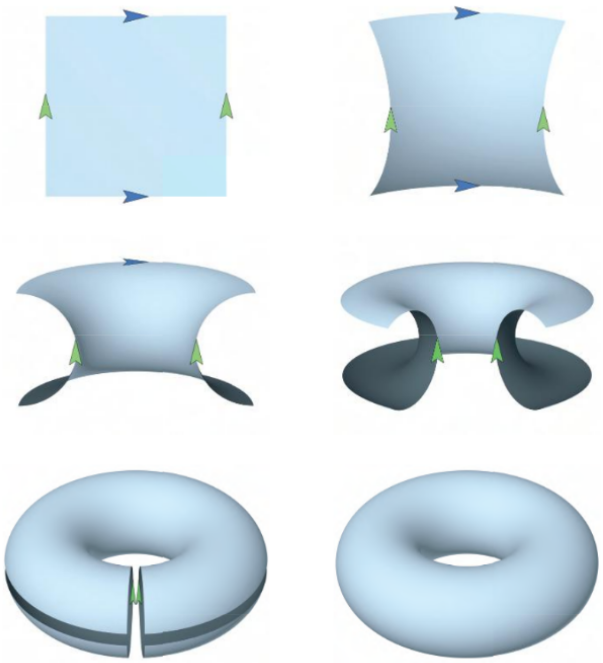


Figure 3: Obtaining the fundamental polygon from the torus (and vice versa) [Wee20, p. 27]

It is often easier to work with the polygon than with the surface or its polyhedral structure. Ever since Henry Roy Brahana (1895-1972), who was the first to rigorously prove that every connected compact surface is topologically equivalent

either to the sphere, or to a connected sum of tori (if orientable) or a connected sum of real projective planes (if non-orientable), most proofs of theorems classifying surfaces have used polygons instead of polyhedra, via the described "cutting and gluing" technique. The fundamental polygon gives a very convenient calculation of the Euler characteristic χ and shows that it is related to the number of holes g, a relationship expressed by the formula $\chi = 2 - 2g$. The number of holes determines the surface more directly than the Euler characteristic χ, but χ carries more information about the geometric properties of the surface, which is more useful in the appropriate context [Sti10, pp. 471–473].

1.4 Surfaces Equipped with Geometry

Here we will build on the previous section, in which we spoke about fundamental polygons, and we will see how, using fundamental polygons, we can "cover" the Euclidean and hyperbolic plane, which will eventually make it possible to equip surfaces with the appropriate geometry.

Definition. A map $\varphi : \tilde{S} \to S$, of a surface $\tilde{S}$ onto S, is called a *covering* if φ is a local homeomorphism (i.e., if φ behaves like a homeomorphism when φ is restricted to a sufficiently small domain).

The first interesting examples of coverings are those of orientable surfaces of genus ≥ 1 (i.e. $\chi \leq 0$), and all these surfaces can be covered by planes. The essence of this idea comes from Hermann Schwarz, and it became known through Klein's letter to Poincaré (1882).

The Case genus $= 1$

Before we explain the construction of the universal cover, let us look at how the fundamental polygon of the torus touches itself on the torus. In Figure 4, the torus is shown along with its associated square as the fundamental polygon F on the right.

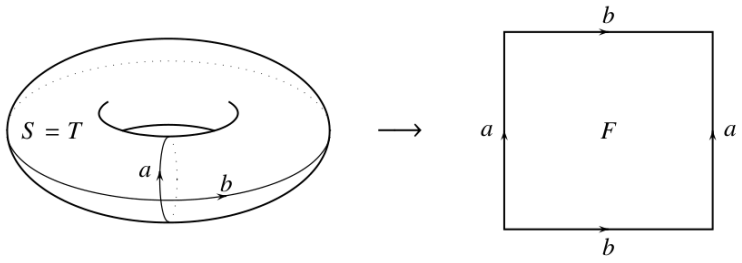


Figure 4: Fundamental polygon of the torus [Sti10, p. 481]

On the torus we can observe (see also Figure 3) that the sides of the square a, b are identified with themselves (in the same direction, as indicated by the arrows on the sides).

Now, the construction of the universal cover of the surface S is carried out by arranging infinitely many copies of the fundamental polygon F of the surface S in the plane, so that neighboring copies of F touch each other in the same way that F touches *itself* on S, as we can see in the following illustration (Figure 5, [Sti10, p. 482]). If we now consider the map $\varphi : \tilde{T} \to T$, where $\tilde{T}$ is the surface

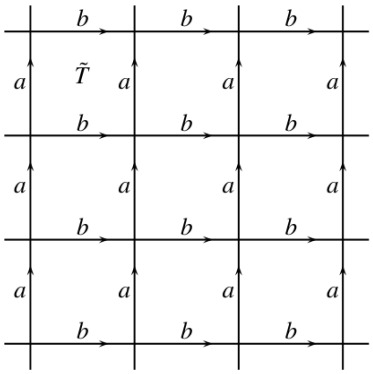


Figure 5: Tessellation of the cover

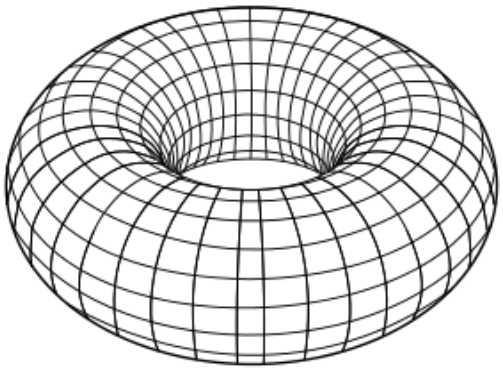


Figure 6: Torus equipped with geometry

obtained by joining infinitely many polygons in the appropriate way (we will soon see that this surface can be regarded as the plane), and T is the torus, we obtain the universal cover φ by mapping each copy of the polygon F in $\tilde{T}$ onto T in the natural way.

The tessellation in Figure 5 can be realized by squares in the Euclidean plane, so we can equip the torus with Euclidean geometry (Figure 6 [Wik26]), where for the distance between two points on the torus we take the distance between their preimages in the plane. The geodesics on the torus are then in fact images of straight lines in $\mathbb{R}^2$. In this way the geometry of the torus is Euclidean, but only locally; globally it is not, because there exist closed geodesics, such as the images of the segments a and b from the fundamental polygon. As a consequence, the sum of the angles of every triangle on the torus is $180°$.

The Case genus > 1

Above we considered the universal cover of a surface of genus 1, i.e., with one hole. However, when the genus is > 1 (i.e., $\chi < 0$), the situation changes drastically, and the natural geometry that now imposes itself is hyperbolic. Namely, according to the Gauss-Bonnet theorem, we know that the curvature of a surface depends on the Euler characteristic χ, and according to the same theorem, surfaces of constant negative curvature satisfy the axioms of hyperbolic geometry. It follows directly that the fundamental polygons of surfaces with two or more holes naturally carry a hyperbolic geometry. Thus the covers will now live in the hyperbolic plane, rather than in the Euclidean one, as was the case for the one-holed torus. We can best observe this in the example of the two-holed torus and its fundamental polygon.

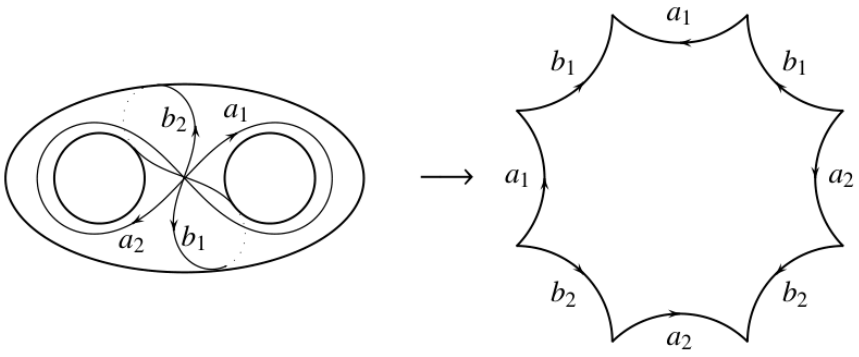


Figure 7: Fundamental polygon of a genus-2 surface [Sti10, p. 483]

We again construct the universal cover so that the polygons touch each other

in the same way that the polygon touches itself on the surface (this can be seen in the figure above, left). Eight copies of the illustrated octagon shown above, right, must meet at every vertex, just as four squares met at every vertex of each square in the previous case. Such a tessellation is impossible with octagons in the Euclidean plane, but it is possible in the hyperbolic plane (Figure 8 [Geo20]).

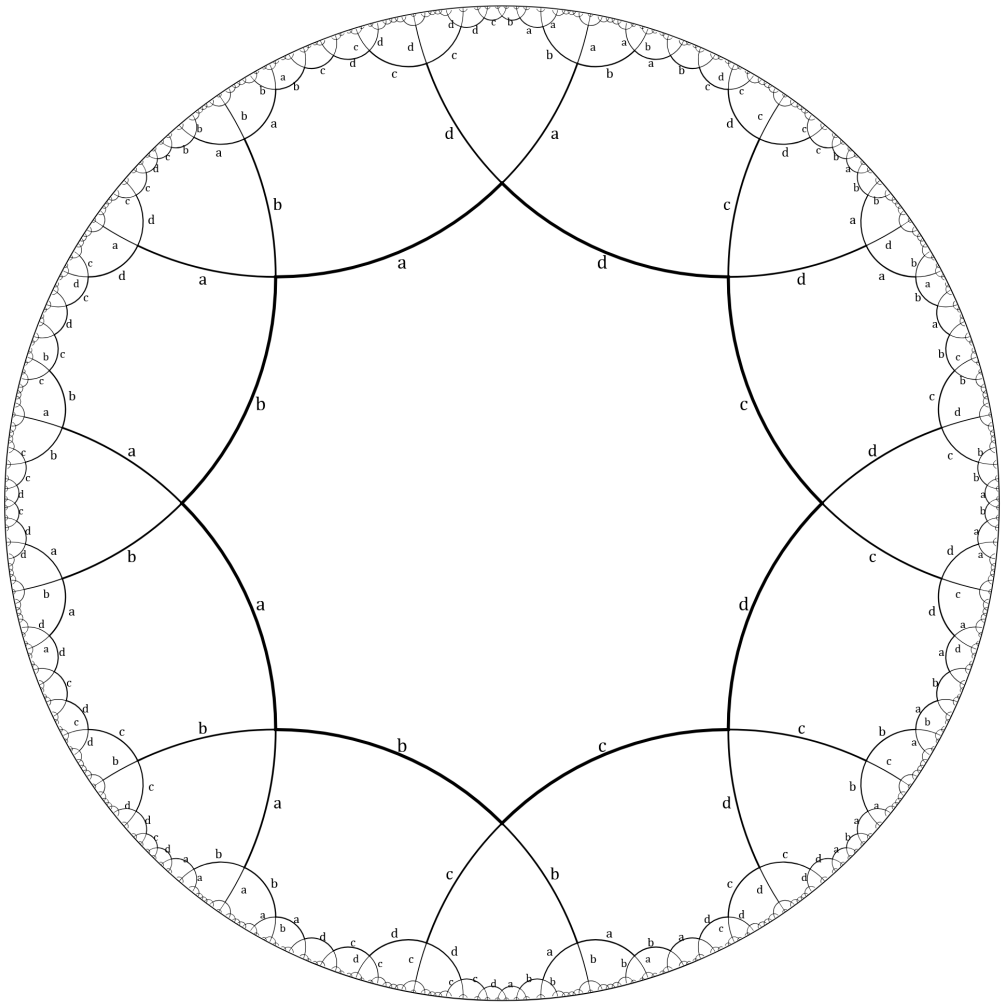


Figure 8: Tessellation of the hyperbolic plane by the fundamental polygon of a genus-2 surface

In general, tessellations by fundamental polygons of surfaces of genus ≥ 2

can similarly be realized in the hyperbolic plane, and were considered by Poincaré (1882) and Klein (1882). As in the case of genus-1 surfaces, surfaces of genus ≥ 2 can also be equipped with a metric and all the rest of their local geometry, only now it will be hyperbolic geometry [Sti10, pp. 479–483].

1.5 The Fundamental Group

Homotopy and Homotopy Classes

Let us now return to the universal cover of the torus (Figure 5). How does a path on the torus correspond to a path in the plane (i.e., in the polygonal tessellation)?

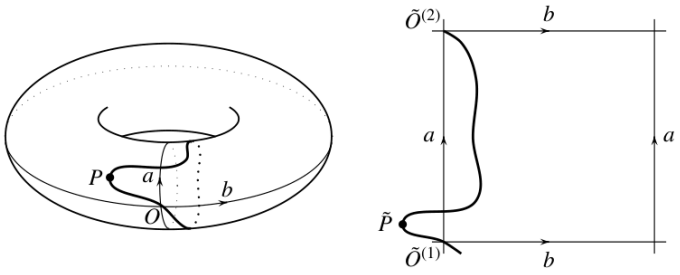


Figure 9: A path on the torus and in the polygonal tessellation (plane) [Sti10, p. 484]

When the point P moves along a path on the torus, the preimage of P, $\tilde{P}$, moves analogously in the plane, with the difference that when P passes from one fundamental polygon to another, the point $\tilde{P}$ passes from one corresponding polygon to another. Thus, even when P returns to the starting point of the path on the torus, $\tilde{P}$ will not necessarily return to the starting point in the plane. We can say that the difference between the final and initial position of the point $\tilde{P}$ in some sense measures how many times the point P winds around the torus. When P winds once around the torus, roughly in the direction $\vec{a}$, $\tilde{P}$ moves from one end of the segment $\vec{a}$ to the other in the plane. If P winds around twice, $\tilde{P}$ moves from one end of the segment $\vec{a}$ of one *polygon* to the end of the segment $\vec{a}$ on the other (next) polygon.

Definition. For paths p and p', with a common starting point on a surface S, we say they are *homotopic* if p can be continuously deformed into p' while O remains fixed.

It turns out that homotopy is an equivalence relation, so that all mutually homotopic paths belong to the same so-called homotopy class.

If we deform the path p, on which the point P lies, into p', then the path $\tilde{p}$, on which the point $\tilde{P}$ lies, will deform into the path $\tilde{p}'$, with the same starting and ending points $\tilde{O^{(1)}}$ and $\tilde{O^{(2)}}$ as $\tilde{p}$. It can be shown that every homotopy class corresponds to a translation of the plane, one that takes the starting point of the path ($\tilde{O^{(1)}}$ in the example above) to the ending point of the path ($\tilde{O^{(2)}}$ in the example above). Thus, homotopy classes correspond to discrete translations of the plane (i.e., of the universal cover), namely up, down, left, right, as we will soon see [Sti10, p. 485].

The Fundamental Group $(\mathcal{H}, c)$

In a semi-formal[5] sense, we can define the fundamental group as $(\mathcal{H}, c)$, where $\mathcal{H}$ is the set of homotopy classes of the surface S, and c is the operation of concatenation (joining together) of paths. It turns out that $\mathcal{H}$ equipped with the operation c is indeed a group.

Let S be the torus as in the previous example; is the fundamental group of the torus then isomorphic to some group? It turns out that it is, namely to the group $\mathbb{Z} \times \mathbb{Z}$, since this group corresponds exactly to the discrete translations of the plane tiled by fundamental squares. Namely, we can interpret the integer ordered pair $(1, 0)$ as shifting the entire plane by one field (square) up, or $(2, 1)$ as 2 up, 1 right, etc. And since homotopy classes are determined by the starting and ending points in the plane, there is a natural correspondence with ordered integer pairs [Šar24].

The fundamental group was first defined by Poincaré in 1895, but his definition was more general, since it included figures whose universal covers were not so obvious. For that reason he did not view it as a group that translates the universal cover, as we illustrated above. But Poincaré had already worked on the motions of tessellations (1882), and in 1902 he succeeds in reconciling that earlier work with the fundamental group, arriving at the interpretation we present above.

The fundamental group will eventually lead to the conclusion that one of the basic problems of topology is unsolvable: given two, "sufficiently well described" figures $\mathcal{F}_1, \mathcal{F}_2$, it is not always possible to say whether they are homeomorphic [Sti10, pp. 484–486].

[5]The fundamental group has a considerably different standard notation, but we choose a simple notation that will be familiar to a reader who has encountered algebra.

1.6 The Poincaré Conjecture

Bounding Curves and Homology

To understand the Poincaré conjecture, we must first understand what curves that bound the surface on which they lie are. Let us first consider the case of the sphere. Notice that every simple closed curve on the sphere bounds a surface that is topologically a disk, even when that curve is the equator, since in that case the upper and lower hemispheres can be ”flattened” into a disk. What if we have a loop like the one shown on the torus in the figure below, on the right? If we were to cut the torus along the loop, we would obtain a hollow cylinder, and there are no points of the surface in that hollow, so the loop shown does not bound any part of the surface, and certainly not a disk [Sti10].

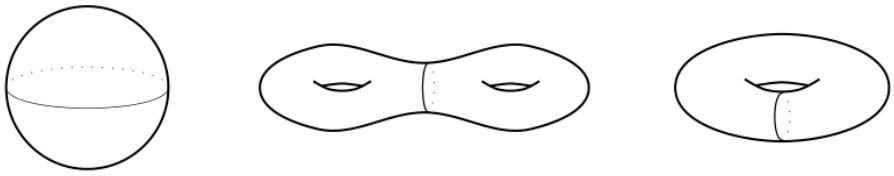


Figure 10: Curves that do and do not bound a surface [Sti10, p. 486]

In the middle figure we have a contour that bounds two surfaces, but these surfaces are not topologically equivalent to disks. Notice that on all three surfaces we can always find contours that bound a disk (by choosing a sufficiently small circle), but on both tori there exist curves that do not bound. Indeed, the sphere $\mathbb{S}^2$ is the only closed surface on which every simple closed contour is bounding. For this reason the sphere is not interesting in this context, whereas the tori are, since for them it makes sense to ask how many curves there are that bound a disk.

This concept also has its own mathematical formalism, called homology, likewise invented by Poincaré, in his work *Analysis Situs* of 1895 and its five supplements. In short, homology is an equivalence relation between loops on a surface, similar to homotopy, except that homotopy induces a stronger kind of equivalence [Šar24].

The Poincaré Conjecture

In 1900, Poincaré conjectured that $\mathbb{S}^3$, the sphere in $\mathbb{R}^4$, is the only closed 3-manifold on which every curve is bounding. Four years later, Poincaré himself discovered a counterexample, the so-called *homology sphere*, which carries this name because this object has the same homology as $\mathbb{S}^3$, despite the two not being homeomorphic. In Poincaré's "homology sphere," every simple closed curve bounds a surface, but not always a disk. Accordingly, Poincaré revised his conjecture to: If every simple closed curve on a closed connected 3-manifold M bounds a disk, then M is homeomorphic to $\mathbb{S}^3$. An equivalent statement of this is that every closed curve on M is contractible, in other words, that it can be shrunk to a point (Figure 11).

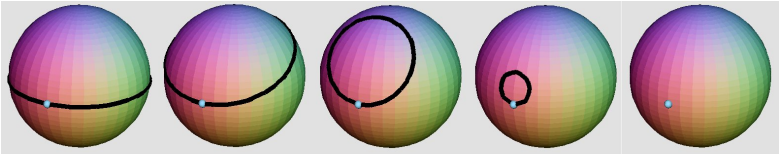


Figure 11: Contractibility of a curve on the sphere [Wik26]

This version would become known as the Poincaré conjecture, one of the greatest mathematical problems of the 20th century. The existence of the homology sphere shows that three dimensions are considerably more complicated than two, but it was not immediately clear how much more complicated. Dehn (1910), Alexander (1919), and Whitehead (1935) would all find numerous examples of anomalies that were unimaginable in the case of 2-manifolds.

In the fifties and sixties of the last century, the first good news arrived. There was still no proof of the conjecture, but progress was being made in higher dimensions. Namely, Smale (1961) showed the analogue of the conjecture for $\mathbb{S}^n$ for $n \geq 5$. It would turn out that it is easier to work, at least in this context, in five dimensions and above than in three, even though three is harder than two (topologists would say that in higher dimensions "there is more room"). The case for dimension 4 would be proved by Freedman (1982), who at the same time proved a number of open problems concerning 4-manifolds. Nevertheless, his approach was unusual, and his contemporaries did not hold out much hope that it could be adapted to the Poincaré conjecture in three dimensions.

In the late seventies, William Thurston developed his *geometrization conjecture*, of which the Poincaré conjecture is a special case. Thurston succeeded in

proving many special cases of his conjecture, but progress in this field would slow down in the early eighties. This did not discourage topologists, since they still wanted a purely topological proof of the Poincaré conjecture, one not relying on geometry. But geometry would nevertheless come to the rescue, namely *differential geometry* [Sti10, pp. 486–489].

The Proof

Below we will present a greatly simplified idea of the proof published in 2003 by Grigori Perelman. The key idea in the proof was to "flow toward homogeneity," an idea originally initiated by Hamilton (1982), where, in the context of 3-manifolds, it goes by the name *Ricci flow*. The difficulties Perelman encountered were numerous, and we will illustrate them using the example of closed manifolds of dimensions 1 and 2, i.e., curves and surfaces. We will call the analogue of Ricci flow in lower dimensions the *pseudo Ricci flow*.

A closed one-dimensional manifold can be realized as a closed smooth curve in the plane, as in Figure 12 (image source: [Sti10, p. 489]).

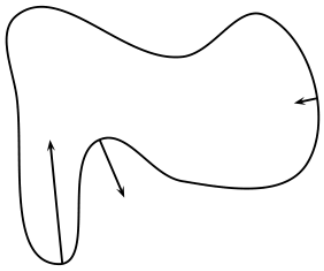


Figure 12: Pseudo Ricci flow of a curve

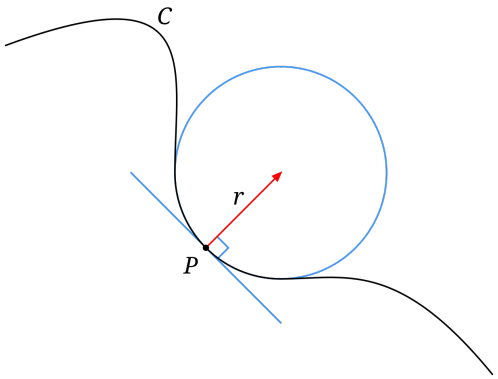


Figure 13: Osculating circle

The vectors shown are perpendicular to the tangents passing through the points at which the vectors touch the curve; their magnitude is proportional to the curvature, i.e., inversely proportional to the osculating circles, by which we mean the circles tangent to the curve at a given point that best approximate the curve there; the vectors also pass through the centers of the osculating circles. Figure 13 ([Wik26]) shows the osculating circle of a curve. The idea now is to let every point of the curve in Figure 12 flow in the direction of the vector based at

it, at a speed equal to the vector's magnitude. Will this curve converge to some curve in the limiting process? It turns out, and can be proved, that it does, namely to a circle. But for this we must additionally impose the assumption that the enclosed area is preserved during the process, since otherwise the curve collapses to a point. Thus, every closed 1-manifold has a geometric realization as a circle [Num14; Sti10].

If we consider a similar process in $\mathbb{R}^3$, a number of complications arise. First of all, instead of tangent vectors, we will now have vectors normal to the tangent planes at points of the manifold. The next problem is that we do not know which kind of curvature to choose, since Gaussian curvature is no longer the only option. We expect that, whatever curvature we choose, every surface (now in the context of $\mathbb{R}^3$, i.e., 2-manifolds) will, under the process analogous to the pseudo Ricci flow described above for curves, transform into a sphere. However, even when we make a good choice of curvature, some pathological surfaces that are homeomorphic to a sphere, such as the one shown in Figure 14, will not be made by the pseudo Ricci flow to "flow" toward a sphere (which has constant curvature) as we would like. This is because the large[6] curvature at the middle of this surface causes that part to thin out faster than the rest of the surface, resulting in a surface whose "neck" becomes ever thinner, so that we do not obtain the sphere $\mathbb{S}^2$ in the geometric sense, since this surface clearly will not have the constant curvature that a sphere has.

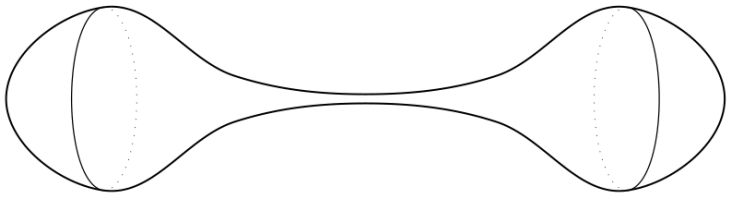


Figure 14: A surface that does not tend toward a sphere [Sti10, p. 490]

The way out of this situation is to perform the so-called surgery: a cut is made on the thin part of the surface, and the resulting holes are then continuously closed up (Figure 15). The resulting surfaces are two surfaces that tend toward the standard sphere under the pseudo Ricci flow process. With curvature analysis, and

[6]Note that this part now has large curvature, which does not match the intuition we might have anticipated from the case of curves

with the help of surgery, one can prove the Poincaré conjecture for 2-manifolds: every closed, simply connected region is homeomorphic to $\mathbb{S}^2$. Of course, this can also be shown more directly, from the topological classification of surfaces. The point is that this method also works for 3-manifolds, where classification theorems are not available.

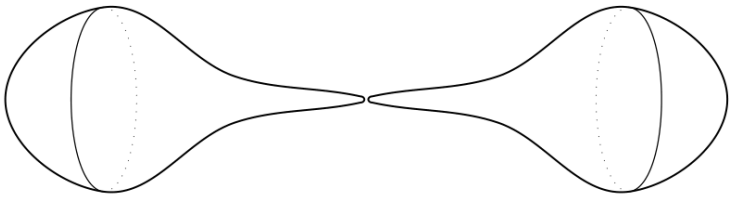


Figure 15: The result of surgery [Sti10, p. 490]

The surface shown in Figure 14 is of course only an analogue of the kinds of 3-manifolds that presented an obstacle to Hamilton in his attempts to prove the conjecture. Perelman overcame all these obstacles in 2003, when he published the proof on the well-known site arxiv.org, in three parts. For this achievement, Perelman was awarded the Fields Medal in 2006, which he declined to accept. The magazine *The New Yorker* published a story covering all the events surrounding the proof of the Poincaré conjecture, and the interested reader can find more information there [Sti10, pp. 489–491].

1.7 Transition

With this, our review of *geometric topology* comes to an end, and in the next part we will deal with *general*[7] *topology*; we will return to topology and geometry, and to algebraic geometry, in the third part of this exposition, when we tell the story of the union of these two streams of topology.

[7] point-set

2 General Topology

2.1 Georg Cantor and the Beginnings of Point-Set Topology

Between 1872 and 1890, Georg Cantor's (1845-1918) investigation of the properties of certain infinite subsets of the real line led him to define the following notions: accumulation point, closed set, derived set (the set of all accumulation points), dense set, etc.

Georg Cantor studied in Berlin under Kummer, Kronecker, and Weierstrass. In 1869 he became a lecturer at the University of Halle. His colleague at the university, Eduard Heine, suggested that he study the uniqueness of the representation of a function by trigonometric series. In 1870 he published a proof of the theorem stating that if the trigonometric series

$$\frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \sin nx + b_n \cos nx)$$

converges to $f(x)$ for all $x \in (0, 2\pi)$, then $f(x)$ cannot be represented by another trigonometric series that converges to $f(x)$ for all $x \in (0, 2\pi)$. A year later, Cantor strengthened this proof, further showing that the representation of the function $f(x)$ is unique even when the series fails to converge to $f(x)$ on a finite set of certain special points [KM99, p. 203].

It was precisely this problem that led Cantor toward the theory of transfinite sets and his results in topology. He realized that the hypotheses of the theorem could be weakened further, namely, that the "special" points are in fact accumulation points.

Definition. A real number x is an accumulation point of a set $X \subset \mathbb{R}$ if for every $\varepsilon > 0$ there exists $y \in X$ such that $0 < |x - y| < \varepsilon$.

Even when the set of accumulation points is infinite, if it itself has only finitely many accumulation points, the theorem still holds. Further work on this question would lead Cantor to his definition of the real numbers, one independent of those of Weierstrass, Dedekind, and others. Using this definition, and the whole accompanying apparatus he developed alongside it, Cantor proved the uniqueness theorem for trigonometric series for "special" sets *of type* v, for $v \in \mathbb{N}$.

Cantor described a closed set as a set that contains its accumulation points, and called the set of accumulation points of a given set its derived set. Through these notions, Cantor "inaugurated" the field we today call point-set, or general, topology, the part of topology closest to set theory and most relevant to analysis [Bur10, p. 729], [KM99, p. 204].

Georg Cantor also stands out as the mathematician who had the greatest influence on popularizing the point-set doctrine, which gave an entirely new perspective to topological thinking. He published his most influential research in this field in a series of six papers, titled "On Infinite, Linear Point-Manifolds"[8] (1879-1883). Ernst Zermelo would later say that these brilliant papers represent the "quintessence of Cantor's life's work."

In these papers too, the fundamental concept was that of the accumulation point. Building on the Bolzano-Weierstrass theorem, Cantor derived all the concepts well known today: the derived set, the everywhere-dense set, the isolated set of points, the nowhere-dense set, and Cantor's "middle-third" "discontinuum." These (and other) notions in fact opened up entirely new fields of study. Analysts were the first to recognize the usefulness of Cantor's ideas, which would significantly speed up the progress of the theory of real and complex functions in the years following the publication of Cantor's papers.

Applications of Cantor's tools in geometry would come somewhat later, and we will mention them when we discuss how Peano and Jordan made use of these very tools in that domain, and how this in turn influenced the development of point-set topology [Cri99, p. 9].

2.2 Fréchet and Metric Spaces

Point-set topologists initially concerned themselves only with Euclidean spaces, from the line through the plane to higher-dimensional Euclidean spaces, even though all the topological concepts they used had a more general validity. This more general validity was recognized, among the first, by Fréchet, who is considered the first to have worked out a theory of topology for abstract spaces, ones that did not necessarily have any connection to the Euclidean spaces exclusively studied up to that point. In 1904 and 1905, Fréchet published a series of short papers laying the foundations for his thesis, which he would defend in 1905, and which is one of the most important contributions to mathematics. In the thesis, and in the aforementioned series of papers, Fréchet abstracted the theory of analysis to abstract spaces; more precisely, he investigated spaces in which accumulation points of sets and sequences can be defined. To determine whether x is an accumulation point of a set X, it is essential to be able to speak of the "nearness" of the point x to the set X. The question of "nearness" leads Fréchet to consider what we today call metric spaces (Hausdorff coined the name "metric

[8] Über unendliche, lineare Punktmannichfaltigkeiten

spaces" in his 1914 book). These are spaces equipped with a structure that makes it possible to speak of the distance between two points of a given space (set). For the newly minted notion of metric spaces to be useful for the mathematics of the time, Fréchet, inspired by the distance in Euclidean spaces, specifically the real line, defined a general notion of the distance between two points x, y, denoted $d(x, y)$, which had to satisfy 4 properties [Bur10, p. 730]:

Definition. *Metric.* Let X be a metric space, on which a function $d : X \times X \to \mathbb{R}$ is defined. We say d is a metric if the following three properties hold:

1. $d(x, y) = 0$ *iff* $x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, y) \leq d(x, z) + d(z, y)$

Thus, by a metric space we mean nothing more than that the set X is equipped with a distance function, which we call a metric.

The most obvious example of a metric space is the set $\mathbb{R}^n$, in which the metric is the well-known square root of the sum of squared differences. Formally, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$; then we define the metric d on $\mathbb{R}^n$ by

$$d(\mathbf{x}, \mathbf{y}) := \sqrt{\sum_{k=1}^{n} (x_i - y_i)^2}$$

A more interesting example is the set of all real continuous functions on the interval $[0, 1]$. For two such functions f, g, we define the metric $d(f, g)$ by

$$d(f, g) := \max\{|f(x) - g(x)| : x \in [0, 1]\}.$$

Once we have a notion of distance between the elements of a set, we can also formalize, in a different way, the idea of an accumulation point x of a set A that is a subset of a space X equipped with a metric d.

Definition. *Accumulation point*, Fréchet's version. x is an accumulation point of a set A if for every $\varepsilon > 0$ there exists a point $y \in A$ such that $y \neq x$ and $d(x, y) < \varepsilon$.

Fréchet also carried over from real analysis the concept of the convergence of a sequence, formulating it so that it depends only on the metric.

Definition. *Convergent sequence.* A sequence $x_1, x_2, x_3, ...$ of points of X converges to x if the sequence of real numbers $d(x_1, x), d(x_2, x), d(x_3, x), \ldots$ converges to 0.

The idea of a continuous function can likewise be expressed in terms of the metric.

Definition. *Continuous function.* Let X, X' be metric spaces, with metrics d, d' respectively; then the function $f : X \to X'$ is continuous at the point x if $d'(f(x_n), f(x)) \to 0$ whenever $d(x_n, x) \to 0$.

This too is the standard adaptation of sequential continuity familiar to us from real analysis [Bur10, p. 731].

2.3 Felix Hausdorff and Neighborhoods

Topology first appears as a cohesive mathematical discipline in Felix Hausdorff's textbook, published in 1914 under the title *Grundzüge der Mengenlehre (Foundations of Set Theory).* This textbook would remain, right up to the end of the 1920s, the main reference for learning topology and set theory, which, as we have been able to see so far, were closely connected. It is in this very textbook that metric spaces first appear under that name.

An interesting fact is that Hausdorff was not a topologist, nor had he published anything on the subject of topology before his popular textbook [Bur10, p. 731].

The distinctive feature of Hausdorff's approach to topology lies in the fact that he emphasized the importance of the notion of the neighborhood of a point. In[9] 1904, when Fréchet wanted to develop a theory in which the derived set would necessarily be closed, he carried on a written correspondence with Jacques Hadamard (1865-1963), in which Hadamard writes [KM99, p. 209]:

> *Would it not be a good idea to start from the notion of neighborhood, rather than from the notion of limiting value?*

Already in 1905, Fréchet would heed Hadamard's advice.

Even before Fréchet began his work on abstract spaces, David Hilbert, in 1902, proposed that the notion of neighborhood was central. But Hilbert would never pursue this line of work further; nevertheless, his opinion testifies to the fact that the notion of neighborhood had already been maturing for some time [KM99, p. 211].

Various attempts to arrive at an adequate definition of an abstract space culminate in the work of Hausdorff (1868-1942). Already in 1902, Hausdorff was giving lectures on set theory in Leipzig, and this is considered to have been the first course on set theory anywhere in Germany. In 1912, Hausdorff was a

[9] "Probably" that year

professor at the University of Bonn, and there he continued to teach the subject of set theory. In chapter 6 of his notes for that course, preserved in the archive of the University of Bonn, he works with *point sets* (”Punktmengen”), which he also calls *neighborhoods* (”Umgebungen”). Below we reproduce the original portion of Hausdorff’s manuscript in which he characterizes this and other notions [KM99, p. 213].

Hausdorff, Manuscript 1912b, para. 6, Archive of the University of Bonn

Point sets on the straight line, in the plane, in space, and in general in n-dimensional space $r = r_n$. A *point* is defined as n real numbers $(x_1, x_2, \ldots, x_n)$, whose coordinates are orthogonal. We define the *distance* between two points as

$$x \cdot y = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2} \geq 0.$$

By the *neighborhood* U_x of a point x we mean the collection of all points y for which $x \cdot y < \rho$ (ρ a positive number; the interior of a "sphere" of radius ρ).

For purposes of illustration, we will as a rule work with the plane $r = r_2$; if the number of dimensions gives rise to any deviations, we will note this specifically. Neighborhoods have the following properties:

(α) Every U_x contains x and is contained in r.

(β) For any two neighborhoods of the same point, either $U'_x \supseteq U_x$ or $U_x \supseteq U'_x$ holds.

(γ) If y lies in U_x, then there exists a neighborhood U_y contained in U_x ($U_x \supseteq U_y$).

(δ) if $x \neq y$, then there exist two neighborhoods U_x, U_y with no point in common: ($\theta(U_x, U_y) = 0$).

The considerations that follow are based only on these properties. They remain valid in a very general setting as well, provided r is a set of points x, and to the points x there correspond point sets U_x with these 4 properties. One such system, for instance, is the following: if we define the neighborhood of x as the system of points where

$$|x_1 - y_1| < \rho, \quad y_2 = x_2;$$

the neighborhood is then a horizontal segment (without endpoints) of length 2ρ. Or: if we define as a neighborhood the system

$$|x_1 - y_1| < \rho, \quad |x_2 - y_2| < \rho,$$

essentially the interior of a square with side length 2ρ, centered at x, etc.

According to some, it was the logical analysis of the foundations of complex analysis that led Hausdorff to the four neighborhood axioms. In 1914, the aforementioned Hausdorff textbook, *Foundations of Set Theory*, appears. Cantor's works had often blended point-set theory with abstract considerations, whereas in this work Hausdorff carefully draws a distinction between general set theory and point-set theory. In chapter 7, he briefly discusses three possible approaches to turning a set — regarded merely as a system of its elements, without considering the relations between the elements — into a space. His goal was to define a general notion of space encompassing not only $\mathbb{R}^n$, but also Riemann surfaces, infinite-dimensional spaces, and spaces whose elements are curves or functions. As motivation for this he gives two arguments: a significant simplification of the theory, and preventing intuition from leading us astray. The three possible approaches differ in the choice of primitive notions, namely distance, limit, and neighborhood. Which of these three spatial arrangements is chosen, Hausdorff regards as a matter of personal preference, but he also explains what is gained and what is lost with each of the three approaches [KM99, p. 214].

Before stating the four axioms we have already seen in the manuscript, Hausdorff characterizes a topological space as follows:

> *A topological space E is a set to whose elements (points) certain subsets U_x are assigned, which we call neighborhoods of x, in accordance with the following axioms [...].*

Hausdorff's generalization of the notion of space is one of the major contributions to the unification of mathematics. Geometry and analysis had been separate disciplines, and axiomatization put an end to that. Hausdorff succeeded in producing an axiomatization general enough to describe abstract spaces, yet restrictive enough to yield an interesting theory. His theory of topological and metric spaces encompassed all previously existing results and generated new notions and theorems [KM99, p. 215].

All this would contribute to the flourishing of topology in the 1920s and 1930s, and many mathematicians contributed to the field by building on Hausdorff's ideas. A notable example is the duo of P. Alexandroff and Paul Urysohn. Among other things, they introduced the definition of a compact set as we know it today. They further showed that a metric space is compact if and only if every infinite subset contains an accumulation point (the so-called Bolzano-Weierstrass property). The same duo also gave the main results on the metrization problem for topological spaces. Namely, although every metric space is a topological

space, there exist topological spaces whose neighborhoods cannot be specified by any metric. The metrization problem was to find the conditions under which a topological space could be regarded as metric, and it would not be fully solved until the early 1950s [Bur10, p. 732].

2.4 Transition

With this we conclude our overview of general topology. We can already sense that, in the years following Hausdorff, geometry and analysis, and indeed the two ”kinds” of topology, would become ever more connected, culminating in their unification, about which we will say more in the next section.

3 Synthesis

In this part we will see how, over time, geometric-combinatorial topology and general (point-set) topology came to form a single discipline. We will present the process of synthesis through the development of several key concepts, through which these two historical lines in the development of topology became interwoven:

- dimension
- curves and pathological examples of curves
- manifold

3.1 Dimension

The First Symptoms

In a letter dated January 5, 1874, Cantor writes to Dedekind:

> *Can a surface (say, a square together with its boundary) be put into bijective correspondence with a line (or with a segment including its endpoints), so that to each point of the surface there corresponds a point of the line, and conversely, so that to each point of the line there corresponds a point of the surface?*

Cantor was aware of the importance and difficulty of this question, and understood that most mathematicians would dismiss the possibility of such a correspondence as something so obvious that no proof was even needed. Cantor would work on this problem, most likely intermittently, from 1874 until April 1877, but without success. On the anniversary of Gauss's 100th birthday, Cantor mentioned this problem in conversation with colleagues (among others, Heinrich Weber and Rudolf Lipschitz), to which most replied that a bijection between objects of different dimension was obviously impossible. Nevertheless, Cantor insisted on a proof.

Soon afterward, Cantor changed his approach, namely he now tried to construct the desired bijection. This would soon prove fruitful. In 1877, Georg Cantor arrived at an incredible discovery: there is a bijection between the segment $[0, 1]$ and the square $[0, 1] \times [0, 1]$, and moreover, also between the unit segment and the n-dimensional cube. He quickly reported this to his friend Dedekind [Cri99, pp. 4–5]:

As much as you may disagree with me, I can only say: I see it, but I don't believe it. ... Surfaces, solids, and even continuous figures of ρ dimensions can be placed in bijective correspondence with continuous lines, that is, with figures of only one dimension; ...

The implications of this result for the notion of dimension, which geometers had used intuitively and informally, were clear to Cantor. For example, geometers regarded the infinitude of a surface as the square of the infinitude of a line, or the infinitude of a cube as the cube of the infinitude of a line. It therefore made sense to ask whether the intuitive notion of dimension could be well defined, and whether it had any meaning at all.

It is important to emphasize that Cantor approached the concept of dimension from a non-geometric standpoint. His work was based on bijections and cardinality, that is, from the point of view of set theory, which he was creating at precisely that time.

Before Cantor, mathematicians such as Dedekind, Riemann, and Helmholtz had an informal conception of a manifold, one that depended on the number of coordinates. In other words, the notion of dimension was closely tied to the number of coordinates. In yet another letter to Dedekind in June 1877, Cantor would express the view that "...the distinction between manifolds of different dimensions must be sought in other aspects, entirely different from the number of coordinates..."

Dedekind, although convinced of Cantor's results, did not agree that the foundations of geometry had been undermined, as Cantor was inclined to claim. He saw a way out of the situation through continuity, which Riemann and Helmholtz had presupposed, but with questionable explicitness. Nevertheless, it became necessary to re-examine the notion of dimension rigorously.

Dedekind himself realized that a theorem on the invariance of dimension was needed, one that would incorporate continuity. In a letter to Cantor in July 1877, he succeeded in formulating the problem, and the following statement, very clearly — a statement that is essentially the invariance theorem:

If a bijective and complete correspondence can be established between the points of a continuous manifold A of a dimensions, on the one hand, and a continuous manifold B of b dimensions, on the other, then this correspondence must necessarily be discontinuous if a and b are different.

It would turn out that this theorem was easier to formulate than to prove, but it was important that the sensitive points of the notion of dimension had been located.

It is worth noting that some publishers of mathematical journals initially regarded Cantor's result as bizarre, and at first refused to publish his papers [Cri99, pp. 5–7].

Dimension as an Invariant

At the start of 1878, Cantor published "A Contribution to the Theory of Manifolds," which set off a burst of mathematical activity. The goal was to salvage the notion of dimension; the paradoxical correspondence between manifolds of different dimensions had to be explained. Five mathematicians, around the same time as the publication of Cantor's paper, attempted to demonstrate the invariance of dimension, but owing to the immaturity of topology at the time, they were confined to using methods of real analysis and elementary geometry, which would prove insufficient for a result of the desired generality.

Among these attempts, we single out that of Eugen Netto (1848-1919) from 1878, which interested Cantor the most, for the reason that it seemed the best attempt to prove the invariance theorem in full generality, and because it convincingly placed the question of dimension within the domain of topology. Nevertheless, not even this proof fully convinced Cantor that the matter was settled, as we know from his letters to Dedekind. Thus Cantor himself would attempt to give his own proof in 1879, one of an inductive nature, in which he demonstrated a contradiction with the mean value theorem. Cantor's proof would eventually be refuted by Peano's innovation of the space-filling curve, which would not be published until eleven years later, by which time the situation in topology and dimension theory would be drastically changed.

Nevertheless, none of these shortcomings affected mathematicians' sentiment about dimension in the last two decades of the nineteenth century: most were under the impression that the invariance of dimension had been rigorously established [Cri99, pp. 8–9].

3.2 Peano, Jordan, and Their Curves

Mathematical rivals in many respects, Giuseppe Peano (1858-1932) and Camille Jordan (1838-1922) were the first to notice the potential and importance of Cantor's topological ideas for intuitive geometric ideas.

Among topologists, Jordan is best known for his work on the theory of closed curves in the plane. Like Cantor, he insisted on proving what seemed obvious to others, namely that a simple closed curve divides the plane into an interior and an exterior part. This led him to formulate and prove the following theorem [Cri99, p. 10].

> **Theorem.** *Every continuous curve C divides the plane into two regions, an exterior and an interior one; the latter cannot be reduced to zero, since it contains a circle of finite diameter.*

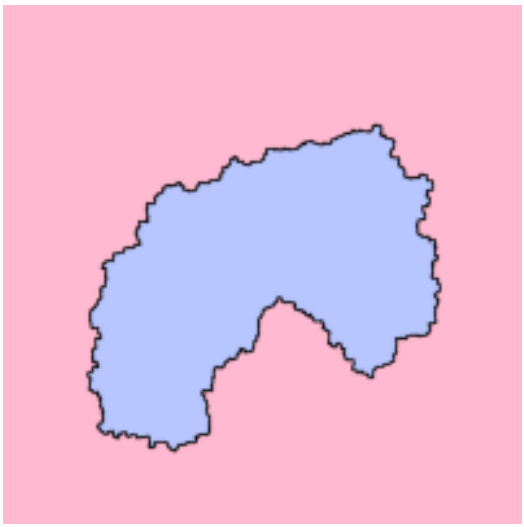


Figure 16: A curve dividing the plane into an interior and an exterior part [Wik26]

Peano, on the other hand, discovered in 1890 a fascinating kind of curve: a space-filling curve. Before that, Peano had worked in analysis and on applying analysis to geometry, in the course of which he also defined various notions such as interior point, exterior point, and boundary point. It is speculated that the set-theoretic approach inspired him to think about counterintuitive situations, which is what resulted in his spectacular curve.

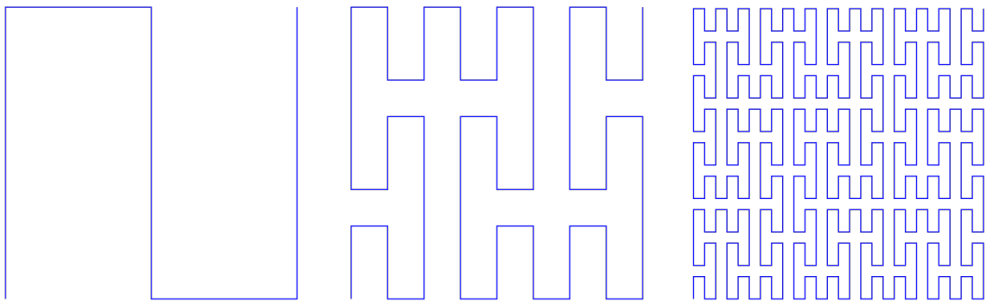


Figure 17: Three iterations of the construction of Peano's curve. In the limiting process, the curve tends toward a filled square [Wik26].

Cantor, Peano, and Jordan were so productive that this created a critical mass of papers, which motivated a good number of mathematicians to recognize the need for deeper research in general topology [Cri99, p. 10].

3.3 Hurwitz and His Program

One of these mathematicians was Adolf Hurwitz (1859-1919). At the First International Congress of Mathematicians, held in Zurich in 1897, he sketched out a plan for further research within the framework of set-theoretic topology. In his lecture, Hurwitz reviewed the progress made in the theory of analytic functions. Specifically, he tried to determine precisely which kinds of curves were suitable for use with Cauchy's integral theorem. In his efforts to deal with these problems, Hurwitz placed them in the context of the topology of closed sets.

His treatment was, in many respects, reminiscent of Cantor's work on cardinal and ordinal numbers — in the sense that closed sets could be grouped (assigned to classes) so that a single class contains sets that are mutually in bijective correspondence, sets he called "equivalent." Hurwitz would then present the following goal:

> The goal of *analysis situs*[10] is to seek invariants of classes of point sets.

Hurwitz also gives an example: Jordan's simple closed curve is a set of points of the plane belonging to the class of sets that contain the boundary of a square. In other words, with his program Hurwitz placed the central problem of seeking

[10]The name topology carried during that period

topological invariants within the context of point-set theory, one of the first and most concrete steps in clearing a path between geometric and general topology.

Thus the set theory of the time became a nest for pathological geometric examples. An early catalogue of unusual curves can be found in the first book on set theory written in English: *The Theory of Sets of Points*, by William H. Young and Grace Chisholm Young (1906). For mathematicians well trained in Cantor's methods, topological monsters became tolerable and less alien. They remained off-putting to those who tried to pursue "normal mathematics," but these pathological oddities were important for the development of general topology and its penetration into other spheres of mathematics [Cri99, pp. 10, 11].

3.4 Hausdorff and the Initiation of the Synthesis

The formal reconciliation of geometric-combinatorial and general topology reaches its culmination with Hausdorff's *Foundations of Set Theory*, published in 1914, which we have already mentioned. It is worth noting that this textbook was renowned for being exceptionally readable.

In his book, Felix Hausdorff presented a synthesis of various results of point-set theory through his theory of topological spaces. He combined fundamental geometric ideas with the abstract analytic concepts of David Hilbert, Maurice Fréchet, and Hermann Weyl.

Still following in Bolzano's footsteps and his insistence on giving precise definitions to geometric objects, mathematicians, faced with the emergence of topological monsters, were driven into a corner where they had to put an end to intuitive definitions. This would lead Hausdorff to remark, pessimistically:

> *We have not given a definition of the concept of curve; the sets that bear this name by convention are of such a heterogeneous nature that they do not fall under any reasonable common concept.*

Although everyone understood the notion of a curve well intuitively, no one could adequately define the general concept of a curve. All attempts were either too broad (Jordan) and included pathologies, or too narrow (Nels Lennes) and excluded important examples of curves [Cri99, pp. 18, 19].

3.5 Brouwer's Contribution, the Return to Dimension, and the Beginnings of Algebraic Topology

Luitzen Egbertus Jan Brouwer (1881-1966) had a major influence on the emergence of the new topology in the early years of the twentieth century. He had a rather different view of topology than Poincaré. Working within the framework of set theory and general topology, Brouwer developed topology to new limits, with his theory of mappings, degree, and dimension. He wrote his main papers in topology between 1909 and 1913.

Although he had already published papers in classical geometry, his doctoral thesis, *On the Foundations of Mathematics* (1907), marks the true beginning of his mathematical career. This work revealed a twofold interest in mathematics that would remain visible throughout his entire career: his concern for the critical assessment of the foundations of mathematics, which would lead him to create

intuitionism, and his deep interest in geometry, on account of which he made a very significant contribution to topology [Cri99, p. 14].

Driven by great ambition, Brouwer attacked Hilbert's fifth problem, which led him to engage with Schoenflies's *analysis situs*. Schoenflies was a follower of Hurwitz's program, and was known for his proof of the converse of the Jordan theorem, as well as for his attempts to describe the topology of the plane and to achieve some of the other goals Hurwitz had outlined. In 1909, Brouwer found serious problems in Schoenflies's work, about which he also wrote to Hilbert. By the time Brouwer had finished repairing Schoenflies's work, the only thing that survived was the aforementioned converse of the Jordan theorem, while most of the theory of curves was left in ruins. In particular, Brouwer gave a counterexample showing the inadequacy of Schoenflies's definition of a curve.

Although Brouwer's work exposed the weaknesses of Schoenflies's analysis, Schoenflies's work served Brouwer as a springboard — a completely different situation from Cantor's, who, thirty years earlier, had often started from nothing.

In 1909, at the age of 28, Brouwer presented his program "The Nature of Geometry" for research in topology, in his inaugural lecture as a lecturer at the University of Amsterdam. The study of groups of bijective transformations and of point sets in the plane and in higher-dimensional spaces led him toward *analysis situs*[11]. The need to classify these groups led him to difficult problems in topology, which is one of the earliest witnesses to the connection between abstract algebra and topology [Cri99, p. 15].

[11]The name topology carried at the time



Figure 18: From left to right: P. Alexandroff, Brouwer, P. Urysohn (from the Alexandroff archive) [Cri99, p. 19].

Brouwer was also fascinated by the invariance-of-dimension theorem, which we have already discussed. On January 1, 1910, Brouwer sent a letter to David Hilbert, sketching the main ideas of his influential forthcoming papers. Hans Freudenthal would eventually come to describe these papers as "the cradle of modern topology." Already in March of the same year, he wrote to Hilbert again, this time saying that he had a partial proof of the invariance-of-dimension theorem. In June, Brouwer published "Proof of the Invariance of the Number of Dimensions," which the following month was accompanied by another masterpiece, "On the Mapping of Manifolds," in which he supplemented some ideas from the first paper.

The proof swept away all previous attempts to prove dimensional invariance. The lemma on which the proof depended, Brouwer proved using his innovative methods, such as the degree of a mapping, simplicial decomposition, and approximation of mappings. From this lemma, it was straightforward to prove the following two theorems:

> **Theorem 1.** *An m-dimensional manifold cannot contain the image of a continuous bijective mapping of a domain of higher dimension.*

> **Theorem 2.** *In an m-dimensional manifold, the image of a continuous bijective mapping of a domain of lower dimension is a nowhere-dense set of points.*

Thus, topological mappings between manifolds can neither lower nor raise dimension. Freudenthal would say that this paper marked

> *... the beginning of a new paradigm and of promising methods, today known as algebraic topology. It demonstrates the ideas of simplicial mapping, barycentric extensions, simplicial approximation, small modifications, and, implicitly, the degree of a mapping and its invariance under homotopic change, along with the concept of the homotopy class. It was simply magic.*

These papers are clear today, but in their own time they were difficult to understand, precisely because of all the new ideas. Their influence would become clear and effective only several years later. Indeed, they were fundamental to the later algebraic theories of topology [Cri99, p. 16].

The fundamental group had already been studied intensively, but in these early years it was hard to find results belonging to homotopy theory, until the appearance of Brouwer's papers, published between 1910 and 1912, which heralded a new era of topology. Compared with Poincaré, his expositions were very rigorous. He also avoided homology theory, and yet Brouwer rightly deserves to be regarded as the founder of homotopy theory [Jam99].

After publishing "On the Natural Conception of Dimension" (1913), in which he critically re-examined Poincaré's definition of dimension and offered his own, Brouwer ceased to engage with the mainstream mathematics of his time. His focus shifted to a dedicated study of the foundations of mathematics, and to work on the intuitionistic approach to mathematics, which we discuss at the very end of this paper [Cri99, p. 19].

3.6 On Manifolds

Riemann, Foundations, Geometry

The development of the concept of a manifold in the period between 1850 and 1950 represents a monumental turning point in mathematics. Before the middle of the 19th century, higher-dimensional spaces were largely treated as analytic abstractions rather than as geometric objects. The formal origin of the concept of a manifold lies in Riemann's 1854 lecture, the "Habilitationsvortrag," in which, given the lack of solid mathematical foundations, he was compelled to rely on philosophy. Riemann defines a manifold as a general notion that admits precise special cases. Riemann distinguished between "discrete" and "continuous" manifolds. In this way, Riemann sketched the conceptual beginnings of what would become set theory (discrete manifolds) and topology (continuous manifolds). For the line of development from Riemann through Dedekind to Cantor, and set theory, see the works of Ferreirós (1993, 1996). Riemann proposed that a manifold be characterized by the number of functions (coordinates) needed to locally determine a point, while acknowledging that its global behavior could be complex. This was developed further through the study of Riemann surfaces, where he used "cuts" and connectivity numbers (precursors of Betti numbers) to analyze the topology of a surface via boundary relations [Sch99, pp. 25, 30].

By the 1870s, manifolds were often reduced to so-called "numerical manifolds," that is, subsets of $\mathbb{R}^n$ or of projective space. This simplification helped mathematicians such as Eugenio Beltrami (1835-1900) to interpret non-Euclidean geometry. At first, Beltrami sought a "real basis" for his models through isometric embeddings into Euclidean space, but after encountering Riemann's work he accepted that an n-dimensional manifold could be defined purely as a system of constant curvature, without the need for an embedding [Sch99, pp. 30–34].

The Theoretical Stage and Poincaré's Constructive Approach

At the turn of the 20th century, Henri Poincaré, through his series of revolutionary papers in the field of *analysis situs*, launched a theoretical approach to the investigation and characterization of manifolds of any finite dimension — a fruitful approach of far-reaching scope in the century that followed. Poincaré did not use a formal axiomatic definition of a manifold; instead, he laid out constructive procedures for generating manifolds. He defined manifolds as the zero sets of differentiable functions (using a condition on the rank of the Jacobian), or as a set of locally analytic parametrizations connected via analytic continuation.

Crucially, Poincaré introduced the division of a manifold into cells and the representation of a manifold as a finite geometric *cell complex*, the so-called "polyèdre," which enabled the combinatorial analysis of manifolds. By decomposing manifolds into cells, he developed the calculus of incidence matrices, homology, and torsion invariants, thereby effectively creating the field of combinatorial topology. However, his reliance on intuitive principles such as triangulability and the Hauptvermutung (the existence of a common subdivision) left gaps that later mathematicians spent decades trying to resolve [Sch99, pp. 39–41].

Elaboration and Unification

The twentieth century brought with it the need to reconcile the axiomatic approach (based on neighborhoods) with the constructive approach (i.e., the combinatorial and simplicial techniques of today's algebraic topology). The axiomatic line was begun by David Hilbert in 1902, who sketched a framework for an axiomatic theory of two-dimensional manifolds based on systems of neighborhoods, which Hermann Weyl further elaborated in 1913. Felix Hausdorff, in 1914, provided the necessary logical clarification by introducing the separability axiom, thereby transforming the notion of a manifold into a particular type of topological space. The combinatorial line was developed by Dehn and Heegaard (1907) and by Steinitz (1908), who advocated a purely combinatorial definition in order to avoid the "logical difficulties" of point-set topology. Brouwer connected these two worlds by introducing "measured" simplicial complexes (PL structures), proving the invariance of dimension and of domain via simplicial approximation and the degree of a mapping [Sch99, pp. 46–50].

The Final Synthesis of Approaches to Manifolds

The final synthesis was achieved through the work of Veblen and J.H.C. Whitehead in the period 1931-1932. They defined manifolds by means of a regular groupoid (pseudogroup) of transformations acting on a Hausdorff space. This provided a definitive axiomatic characterization of continuous (TOP), differentiable (C^r), and analytic (C^ω) manifolds. By 1940, it had been shown that combinatorial and axiomatic strategies could coexist for differentiable manifolds. Although Poincaré's intuition was finally confirmed by these rigorous proofs, the 1950s would soon show that this unification is not universal in higher dimensions ($n \geq 4$), where the discovery of exotic structures and non-triangulable manifolds shattered the dream of a simple, uniform structural universe [Sch99, pp. 55–57].

4 Influence on Modern Mathematics

4.1 Algebraic Topology and Category Theory

Samuel Eilenberg (1913-1998) and Norman Steenrod (1910-1971) at one point decided to axiomatize homology and cohomology, the central concepts of algebraic topology. The first axiomatization was foreshadowed in their work *Axiomatic Approach to Homology Theory (1945)*, but it would not be published until 1952, in the celebrated work of the same duo titled *Foundations of Algebraic Topology*, whose main ideas were already known to experts. They developed this theory using the language of category theory, a new field of mathematics that was still in its infancy at the time. This would have a revolutionary effect on mathematical notation, unlike anything seen since the time of Descartes's *La Géométrie* (1637). The diagrams this duo presented not only made algebraic topology more readable, but would go on to spread to other parts of mathematics as well. It is also worth mentioning that the authors here succeeded in clearly and unambiguously separating algebra from topology, and this was part of the historical process by which homological algebra and algebraic topology split apart and emerged as two distinct subjects [May99, p. 666], [BG99, p. 733].

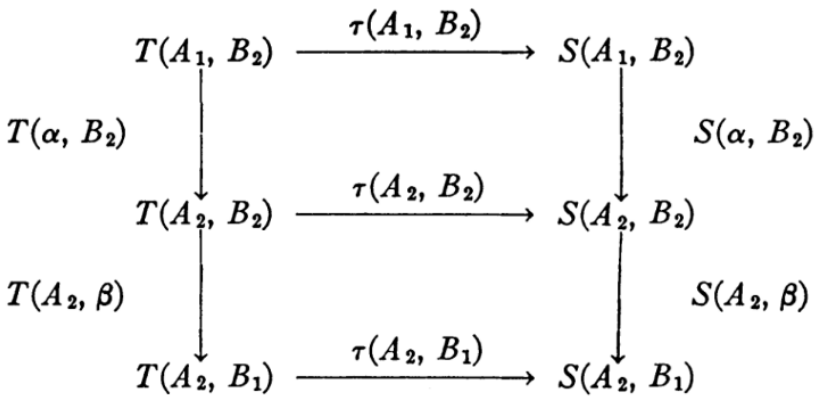


Figure 19: Diagram from *General Theory of Natural Equivalences* (1945)

Officially, categories, together with the key concepts of functors and natural transformations, first appear[12] in that very year, 1945, in the work *General Theory*

[12] Although the work had already been presented in 1942.

of Natural Equivalences by Eilenberg and Saunders Mac Lane (1909-2005). It would turn out that a good deal of mathematics, not only algebraic topology, would not exist at all without category theory [Mar23], [May99].

After the 1945 paper, it was not immediately clear that category theory would become anything more than a convenient language. This was, in fact, the status of the theory for some fifteen years, and it was used in this way both in the aforementioned 1952 work and in Eilenberg and Henri Cartan's (1904-2008) 1956 work on homological algebra. The latter book would crystallize and unify the entire field of homological algebra. These works would enable new generations of mathematicians to learn algebraic topology and homological algebra directly in categorical language, and to master diagrammatic reasoning. Indeed, without the method of "diagram chasing," many results in these works would have been unimaginable, or would at the very least have required a far lengthier presentation [Mar23].

PROPOSITION 2.5. *Every exact sequence* $0 \to A' \to A \to A'' \to 0$ *can be imbedded in a commutative diagram*

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & P' & \longrightarrow & P & \longrightarrow & P'' & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

Figure 20: An example of diagrammatic reasoning from *Homological Algebra (1956)*, Eilenberg, Cartan

The substantiveness of category theory is evidenced by the fact that it made it possible to associate algebraic structures with an object (a topological space), and thereby to obtain topological properties of the object. In algebraic topology,

this association is functorial, meaning that it is not only possible to associate an algebraic structure $A(S)$ with a topological space S, but that considering a mapping from the space S to some other space T naturally leads to a homomorphism between the associated algebraic structures $A(S)$ and $A(T)$. This functorial association of algebraic structures with topological spaces leads to fruitful results about the topological properties of the space in question.

For example, if the requirements on some function $f : X \to Y$ imply incompatible algebraic properties for the morphism $A(f)$, then f cannot exist. Thus, the richer the algebraic structure $A(X)$, the richer the collection of possible results will be [Eyn99, pp. 99, 100].

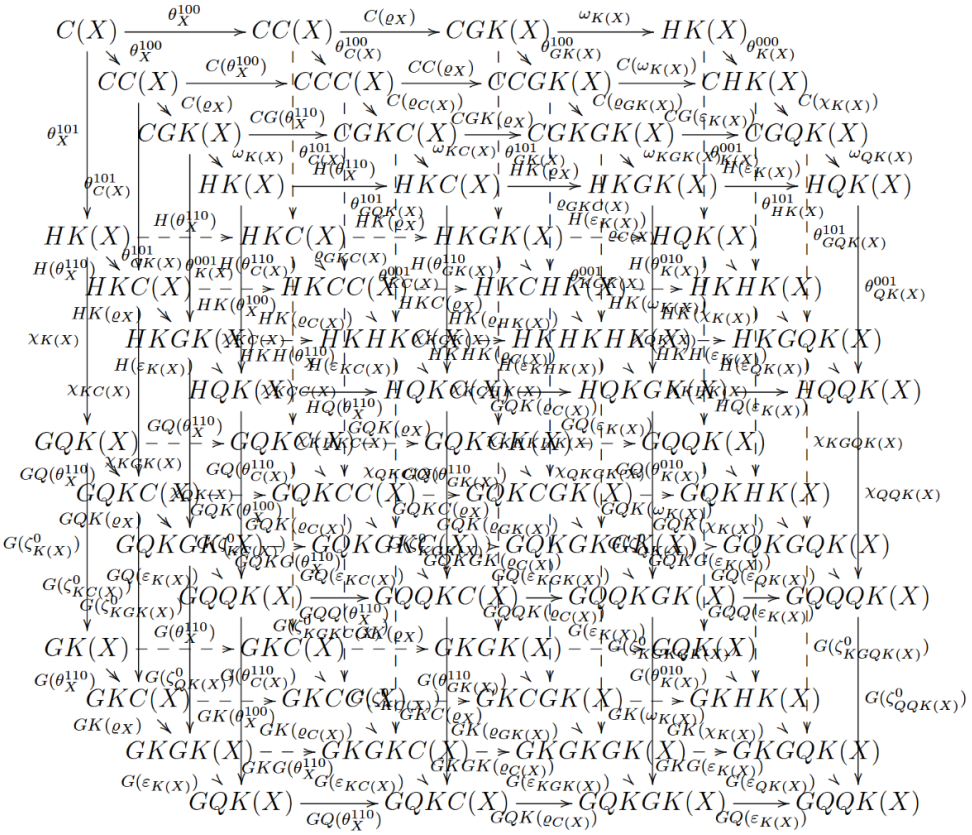


Figure 21: Diagram from modern (higher) category theory (*Comma 2-comonad I: Eilenberg-Moore 2-category of colax coalgebras*, Igor Baković, 2025)

We see that category theory developed hand in hand with algebraic topology,

which testifies to the far-reaching influence of topology on the whole of modern mathematics.

Topos Theory

One of the high points of category theory is the emergence of topos theory, which in a sense returns to its topological roots. Indeed, the striking aspect of this theory is that it unifies two seemingly completely different areas of mathematics: topology and algebraic geometry on one side, and logic and set theory on the other. A so-called *topos*[13] is a category[14] that can be regarded both as a "generalized space" and as a "generalized universe of sets." It is interesting that these different manifestations of the topos developed independently around 1963: the first with Alexander Grothendieck (1928-2014) in his reformulation of sheaf theory for the purposes of algebraic geometry; and the second with William Lawvere (1937-2023) in his quest for an axiomatization of the category of sets, and Paul Cohen in his construction of new models of ZF set theory [LM94, p. 1].

We will also bring toposes into the context of Brouwer's intuitionistic logic, and its connection with topology, which we will briefly touch on below.

4.2 Brouwer's Intuitionism and Topology

Introduction

After a decade of work in topology, in 1913, following the resolution of his work on the notion of dimension, Brouwer withdraws from the front lines of the mathematical research of the time and reorients his interest toward the foundations of mathematics and logic.

The current of intuitionism in the philosophy of mathematics was founded by the Dutch mathematician L.E.J. Brouwer in the early 20th century, and traces its roots back to Brouwer's 1907 thesis. Brouwer claimed that mathematics is the result of purely constructive mental activities of human beings, rather than the discovery of fundamental principles that exist in objective reality. Within this framework, logic becomes subordinate to mathematics, and is regarded as the language by which we store and communicate mathematical patterns, rather than as the foundational system of mathematics. Because in this system truth is equated with *mental constructibility*, intuitionism rejects the law of excluded middle[15], since statements without a concrete construction are not permitted in intuitionism. Consequently, Brouwer proposes that theorems should be divided

[13] "place" in Greek

[14] In the sense of category theory

[15] For infinite domains; for finite ones we can almost always find a constructive proof

into those that are ”true” (constructively provable) and those that are merely ”non-contradictory.”

Out of intuitionism, intuitionistic logic would emerge, whereby the movement reaches its definitive formal stage in the late 1920s through the work of Brouwer's student Arend Heyting (1898-1980) and Andrei Kolmogorov (1903-1987). The formalization and explanation of the meaning of proof in intuitionistic logic, which among other things does not presuppose the law of excluded middle, is set out in the so-called Brouwer-Heyting-Kolmogorov (BHK) interpretation, which is the standard explanation of intuitionistic logic [Att23].

The Connection with Topology, Tarski, Stone

In a topological space, the complement of an open set U is closed but is not usually open, so that among open sets the ”negation” of U should be the interior of its complement. This has the consequence that the double ”negation” of U is not necessarily equal to U. Thus, the algebra of open sets is not Boolean, but instead follows the rules of the intuitionistic propositional calculus, as would be among the first to be noticed by Marshall Harvey Stone (1903-1989) and Alfred Tarski (1901-1983). Since these rules were first formulated by Heyting, we call such an algebra a Heyting algebra. The ”truth values” of any topos form such a Heyting algebra. It would turn out that Stone, Tarski, and also Dana Scott (1932-), with his topological models, foreshadowed a remarkable result: the ”intrinsic” logic of a topos is, in general, intuitionistic. One example of how topos theory incorporates Brouwer's ideas is the topos of real numbers $\mathbb{R}$, in which all functions $\mathbb{R} \to \mathbb{R}$ are continuous [LM94, pp. 5, 6].

In 1937, at the Third Polish Mathematical Congress, Tarski presented his work (The Propositional Calculus and Topology)[16]. Below we give a translation of Tarski's original introduction, which succinctly points to the connection between topology and logic in this context [Tar38].

[16]Original title in German: Aussagenkalkül und Topologie

The Propositional Calculus and Topology

Alfred Tarski (Warsaw)

In this paper I wish to point out certain formal connections between the propositional calculus and topology (and certain other mathematical theories). Above all, this concerns a topological interpretation of two systems of the propositional calculus, namely the ordinary (two-valued, $\top, \bot$) system and the intuitionistic (Brouwer–Heyting) system: to every statement $\mathfrak{A}$ of the propositional calculus there is assigned, in a unique way, a statement $\mathfrak{A}_1$ from topology, such that $\mathfrak{A}$ is provable in the two-valued calculus if and only if $\mathfrak{A}_1$ holds in every topological space; an analogous translation rule is given for the intuitionistic calculus as well. It seems to me that these considerations are of interest not only from a purely formal point of view, but that they also shed light, in an interesting way, on the substantive relationship between the two systems of the propositional calculus and the intuitions underlying them.

Further, in the "Second Main Theorem," Tarski establishes that a logical statement is provable in the intuitionistic calculus if and only if it is valid in every topological space. Under this mapping, logical operations become topological-spatial operations: for example, the negation $\neg A$ corresponds to the *exterior* of a set, i.e., $R \setminus \overline{X}$, while the implication $A \Rightarrow B$ corresponds to the *interior of the union* of the complements of A and B, i.e., $R \setminus \overline{X - Y}$ [Tar38].

The constructivists' rejection of the law of excluded middle is justified, from the topological side, by the existence of the *boundary* of a set. In a topological space, the union of an open set and its exterior covers the whole set only if the set is isolated, whereas in continuous or dense spaces, the boundary represents points that lie neither in the interior nor in the exterior of the set. This boundary formally represents the intuitionistic view that certain statements are neither provably true nor provably false [Tar38; Sto38].

Historically, these discoveries served to illuminate the intuitions of Brouwer and Heyting. Stone observes that whereas classical logic can be represented by operations from set theory, intuitionistic logic requires topological operations in order to be precisely modeled. Once it was proved that the logic of "constructions" is identical to the logic of "open sets," these papers confirmed that topology and intuitionism, in a sense, investigate the same structural reality of mathematical

space [Sto38].

Closing Remarks

From geometry, through analysis, all the way to the very foundations of mathematics, the influences of topology on modern mathematics are numerous and profound. The influences we have discussed are certainly not the only ones; we regret that the story of topology's influence on, among other things, algebraic geometry and graph theory lies beyond the scope of this paper, and that research in this direction would make a fitting complement to this concise overview. We also have not mentioned the work of Johann Benedict Listing (1808-1882), a mathematician who was a pioneer of topology and who coined the term "topology." Finally, one more current and new theory downstream of topology, about which we would have liked to say more, is homotopy type theory (HoTT – *Homotopy Type Theory*), which uses an intuitionistic type theory based on types of objects to which the intuition of homotopy theory – one of the central concepts of algebraic topology – is applicable.

References

[Sto38] Marshall Harvey Stone. "Topological representations of distributive lattices and Brouwerian logics". In: *Časopis pro pěstování matematiky a fysiky 067.1* (1938), pp. 1–25. URL: http://eudml.org/doc/27235.

[Tar38] Alfred Tarski. "Der Aussagenkalkül und die Topologie". In: *Fundamenta Mathematicae 31.1* 24 (1938), pp. 103–134. URL: http://eudml.org/doc/213024.

[LM94] Saunders Mac Lane and Ieke Moerdijk. *Sheaves in Geometry and Logic, A First Introduction to Topos Theory*. Springer, 1994.

[BG99] James C. Becker and Daniel Henry Gottlieb. "A History of Duality in Algebraic Topolog". In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 725–745.

[Cri99] Tony Crilly. "The Emergence of Topological Dimension Theory". In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 1–24.

[Eyn99] Ria Vanden Eynde. "Development of the Concept of Homotopy". In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 65–102.

[Jam99] I.M. James. “From Combinatorial Topology to Algebraic Topology”. In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 561–573.

[KM99] Teun Koetsier and Jan van Mill. “By Their Fruits Ye Shall Know Them: Some Remarks on the Interaction of General Topology with Other Areas of Mathematics”. In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 199–239.

[May99] J.P. May. “Stable Algebraic Topology, 1945-1966”. In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 665–723.

[Sch99] Erhard Scholz. “The Concept of a Manifold, 1850-1950”. In: *History of Topology*. Ed. by I. M. James. Elsevier Science B.V., Amsterdam ; New York, 1999, 1999, pp. 25–64.

[Bur10] David Burton. *The History of Mathematics: An Introduction*. McGraw-Hill Education, 2010.

[Sti10] John Stillwell. *Mathematics and Its History*. Springer, 2010.

[Num14] Numberphile. *Ricci Flow - Numberphile*. 2014. URL: https://www.youtube.com/watch?v=hwOCqA9Xw6A.

[Geo20] Geodavic. *Octagonal Tilings of Hyperbolic Space*. Pristupljeno: 13 Februara 2026. 2020. URL: https://github.com/geodavic/hyperdraw/blob/main/pi2nocolorlabel.pdf.

[Wee20] Jeffrey R. Weeks. *The shape of space*. CRC Press, Boca Raton, 2020.

[Att23] Mark van Atten. *”The Development of Intuitionistic Logic”, The Stanford Encyclopedia of Philosophy (Fall 2023 Edition)*. Edward N. Zalta & Uri Nodelman (eds.), Pristupljeno: 13 Februara 2026. 2023. URL: https://plato.stanford.edu/archives/fall2023/entries/intuitionistic-logic-development/.

[Mar23] Jean-Pierre Marquis. *”Category Theory”, The Stanford Encyclopedia of Philosophy (Fall 2023 Edition)*. Edward N. Zalta & Uri Nodelman (eds.), Pristupljeno: 13 Februara 2026. 2023. URL: https://plato.stanford.edu/archives/fall2023/entries/category-theory/.

[Šar24] Franjo Šarčević. *Kratki uvod u topologiju*. Sarajevo, Zagreb: Univerzitet u Sarajevu - Prirodno-matematički fakultet; Synopsis Zagreb-Sarajevo, 2024.

[Wik26] Wikipedia. *Javno dostupne slike korištene u ovom radu*. Sve slike iz javnog domena. 2026. URL: https://commons.wikimedia.org/.